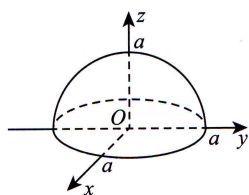


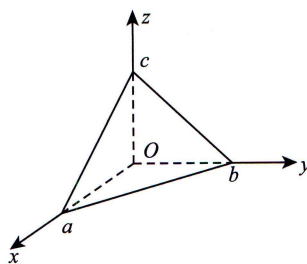
附录3 常用空间图形

(1)



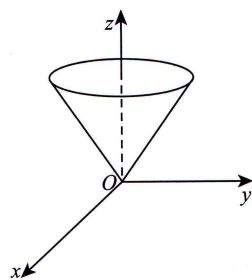
$$z = \sqrt{a^2 - x^2 - y^2}, \quad a > 0$$

(2)



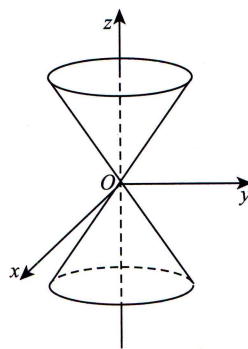
$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1, \quad a, b, c > 0, \\ x, y, z \geq 0$$

(3)



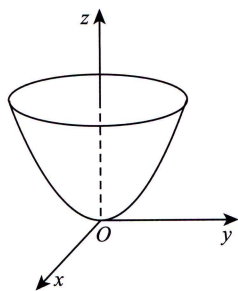
$$z = \sqrt{x^2 + y^2}$$

(4)



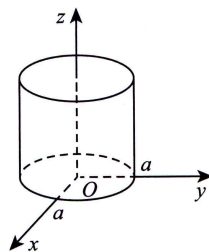
$$x^2 + y^2 = z^2$$

(5)



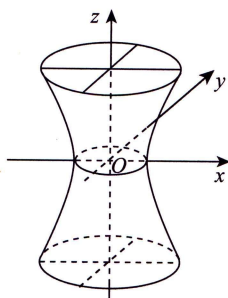
$$z = x^2 + y^2$$

(6)



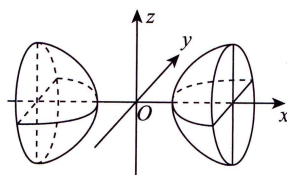
$$x^2 + y^2 = a^2, \quad z \geq 0, \quad a > 0$$

(7)



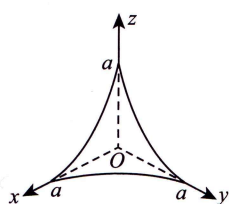
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

(8)



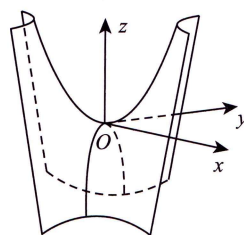
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

(9)



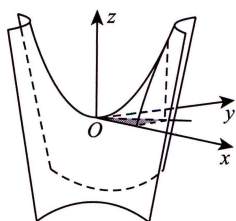
$$\sqrt{x} + \sqrt{y} + \sqrt{z} = \sqrt{a}, \quad a > 0$$

(10)



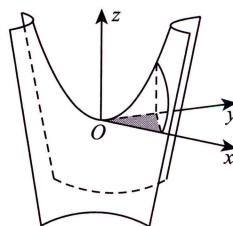
$$z = xy$$

(11)



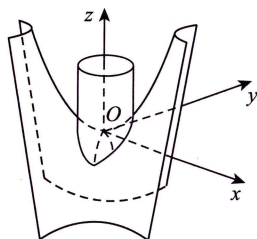
$$\begin{cases} z = xy, \\ y = x, \\ x = 1, \\ z = 0 \end{cases}$$

(12)



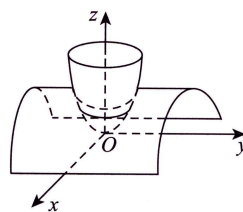
$$\begin{cases} z = xy, \\ x + y = 1, \\ z = 0 \end{cases}$$

(13)



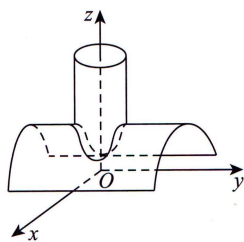
$$\begin{cases} z = xy, \\ x^2 + y^2 = a^2 \quad (a > 0) \end{cases}$$

(14)



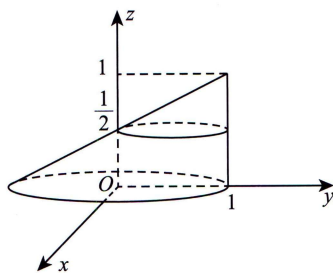
$$\begin{cases} z = x^2 + y^2, \\ z = 1 - x^2 \end{cases}$$

(15)



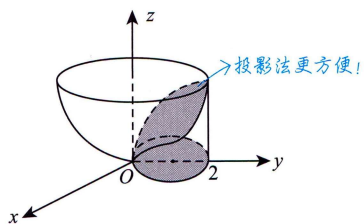
$$\begin{cases} x^2 + y^2 = 1, \\ z = 1 - x^2 \end{cases}$$

(16)



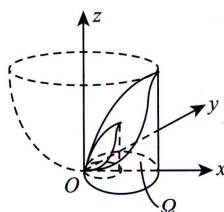
$$x^2 + (y-z)^2 = (1-z)^2, 0 \leq z \leq 1$$

(17)



$$\begin{cases} z = x^2 + y^2, \\ x^2 + (y-1)^2 = 1 \end{cases}$$

(18)



$$\begin{cases} z = 2(x^2 + y^2), \\ x^2 + y^2 = x, \\ x^2 + y^2 = 2x, \\ z = 0 \end{cases}$$